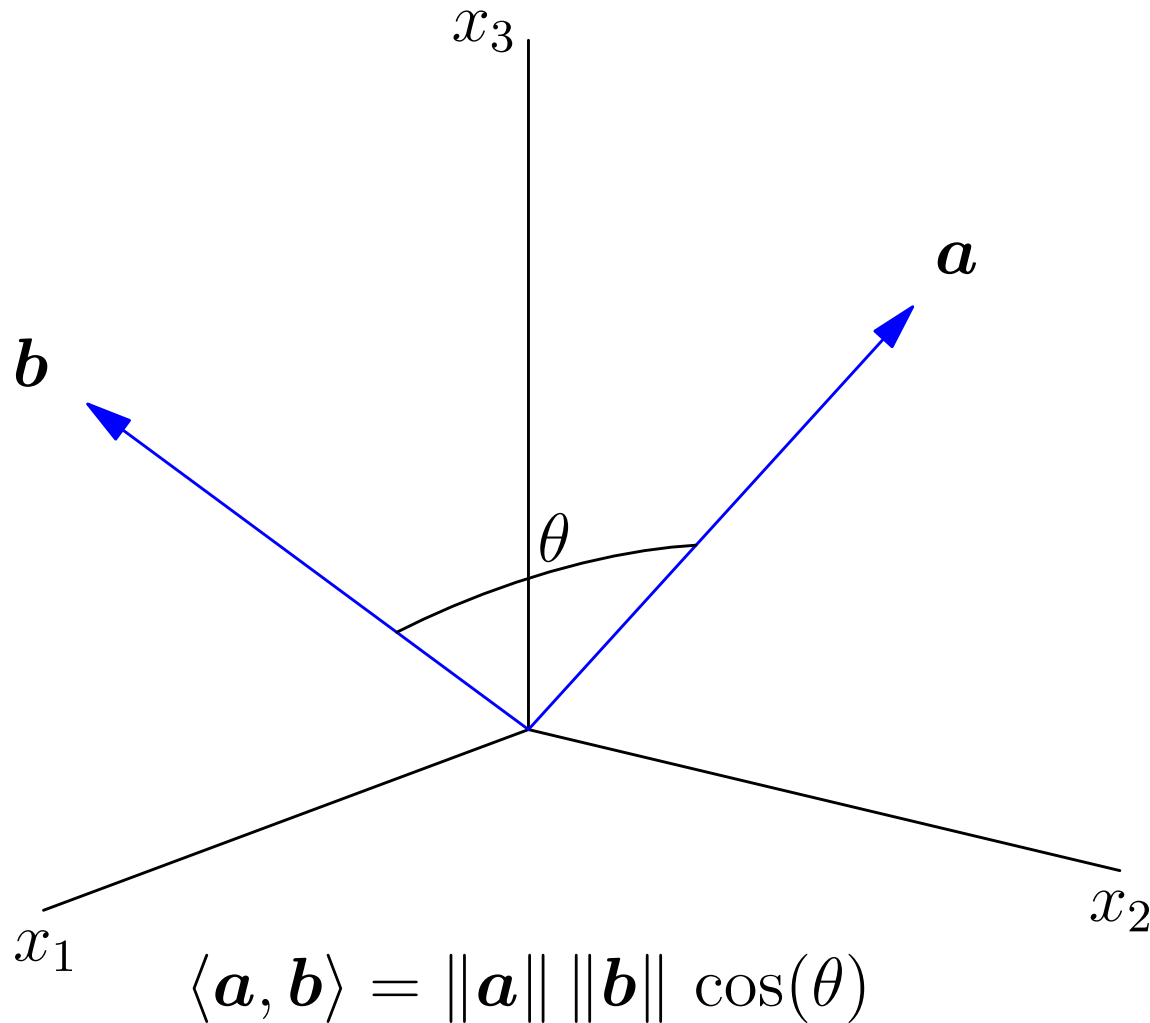


# Mathematics of Machine Learning

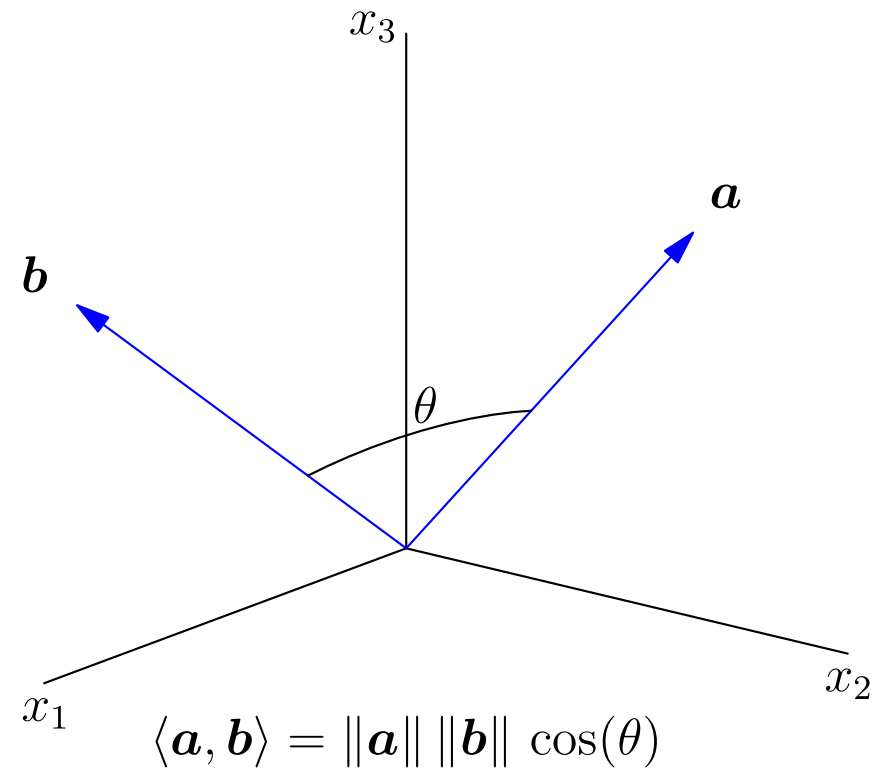
## *Inner Product Spaces*



*Inner products, operators*

# Outline

1. Inner Products
2. Operators



# Recap

- We have looked at vector space
- Recall that vector spaces don't just apply to normal vectors ( $\mathbb{R}^n$ ), but to matrices, functions, sequences, random variables, . . .
- Proper distances or metrics,  $d(\mathbf{x}, \mathbf{y})$ , allow us to construct ideas about geometry of the vector space
- Norms,  $\|\mathbf{x}\|$ , that allow us to reason about the size of vector
- Norm induce a distance,  $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$

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# Inner Products

- We will often consider objects with an *inner product*,  $\langle, \rangle$
- For vectors in  $\mathbb{R}^n$

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y} = \sum_{i=1}^n x_i y_i$$

- For functions

$$\langle f, g \rangle = \int_{x \in \mathcal{I}} f(x) g(x) dx$$

- For  $m \times n$  matrices

$$\langle \mathbf{A}, \mathbf{B} \rangle = \text{Tr} \mathbf{A}^\top \mathbf{B} = \sum_{i=1}^m \sum_{j=1}^n A_{ij} B_{ij}$$

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# Axioms of Inner Products

- An inner product satisfies

1.  $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$  for all  $\mathbf{x} \in \mathcal{V}$
2.  $\langle \mathbf{x}, \mathbf{x} \rangle = 0$  if and only if  $\mathbf{x} = \mathbf{0}$
3.  $\langle \alpha \mathbf{x}, \mathbf{y} \rangle = \alpha \langle \mathbf{x}, \mathbf{y} \rangle$
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5.  $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$

- We can show that  $\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$  satisfies the axioms of a norm, so that an inner-product space is a normed space
- The norm associated with the inner-product for vectors in  $\mathbb{R}^n$  (i.e.  $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y}$ ) is the Euclidean norm  $\|\mathbf{x}\| = \sqrt{\mathbf{x}^\top \mathbf{x}}$

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# Cauchy-Schwarz Inequality

- One of the most important results of inner-product spaces, known as the **Cauchy-Schwarz inequality** is that

$$\langle \mathbf{x}, \mathbf{y} \rangle^2 \leq \langle \mathbf{x}, \mathbf{x} \rangle \langle \mathbf{y}, \mathbf{y} \rangle = \|\mathbf{x}\|^2 \|\mathbf{y}\|^2$$

- Or

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\|$$

- This is a very general result so for example

$$\left| \int f(x)g(x)dx \right| \leq \sqrt{\left( \int f^2(x)dx \right) \left( \int g^2(x)dx \right)}$$

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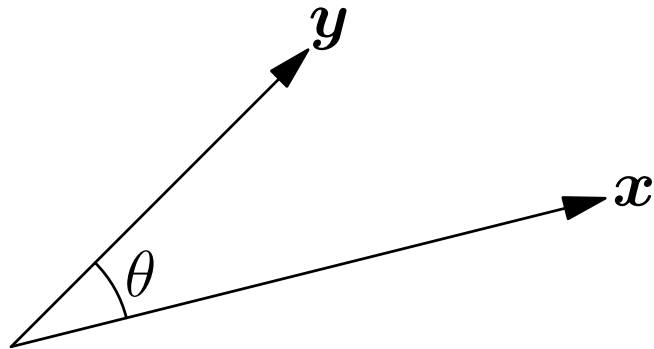
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# Angles Between Vectors

- A natural interpretation of the inner product is in providing a measure of the angle between vectors



$$\langle x, y \rangle = x^T y = \|x\| \|y\| \cos(\theta)$$

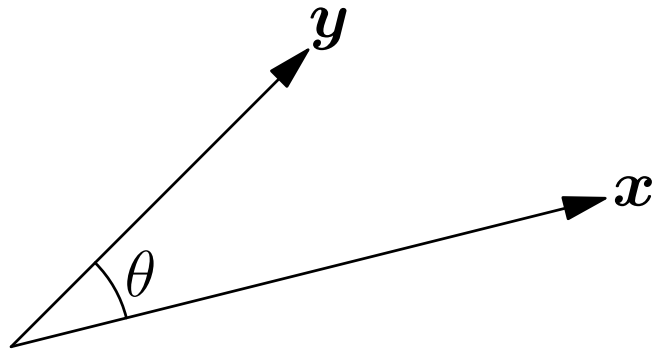
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- We can extend this idea to functions

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- Note that  $\sin(x)$  and  $\cos(x)$  are orthogonal in the interval  $[0, 2\pi]$

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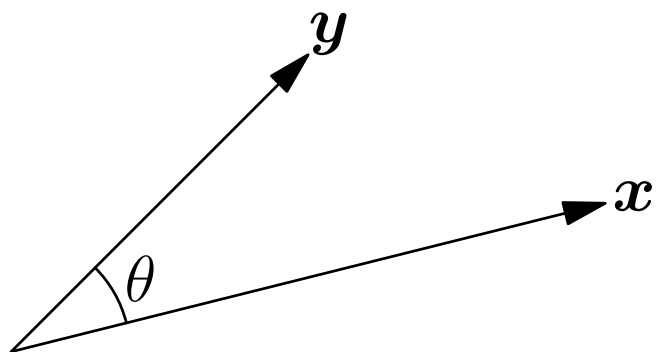
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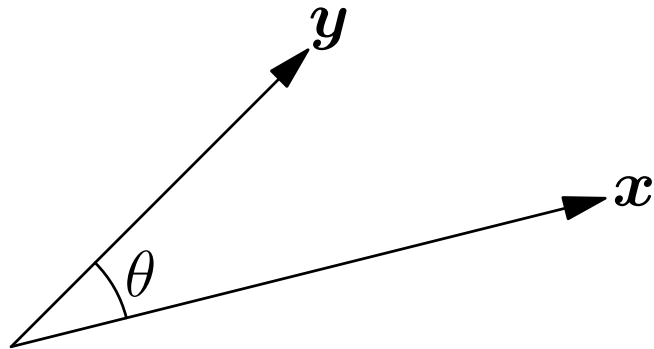
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# Basis Functions

- Any set of vectors  $\{\mathbf{b}_i | i = 1, \dots\}$  that span the space can be used as a basis or coordinate system
- The simplest and most useful case is when the vectors are orthogonal and normalised (i.e.  $\|\mathbf{b}_i\| = 1$ )

- In  $\mathbb{R}^3$  we could use  $\mathbf{b}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ ,  $\mathbf{b}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ ,  $\mathbf{b}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

- This is not unique as we can rotate our basis vectors

- For an orthogonal basis we can write any vector as  $\hat{\mathbf{x}} = \begin{pmatrix} \mathbf{x}^\top \mathbf{b}_1 \\ \mathbf{x}^\top \mathbf{b}_2 \\ \mathbf{x}^\top \mathbf{b}_3 \end{pmatrix}$

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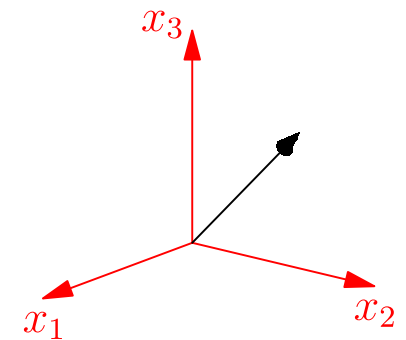
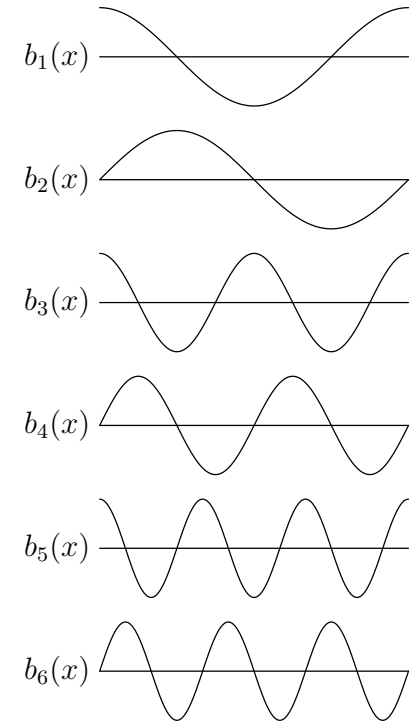
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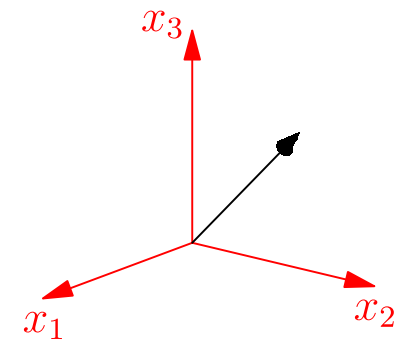
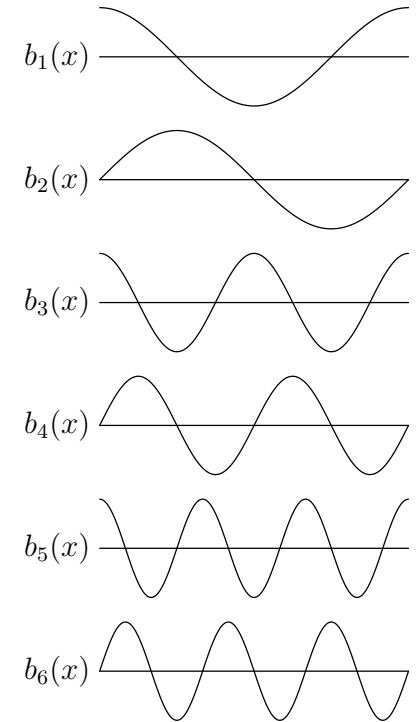
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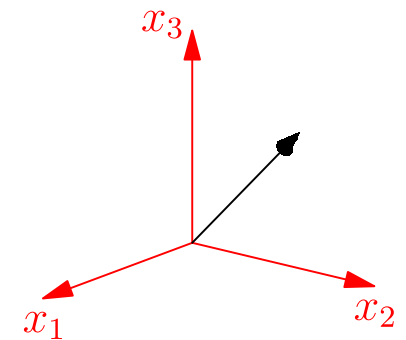
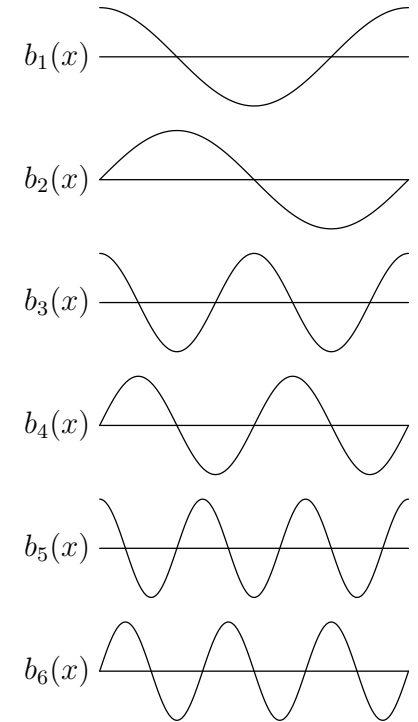
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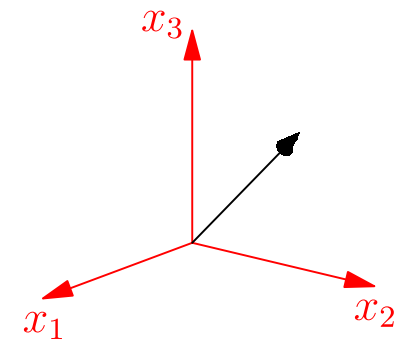
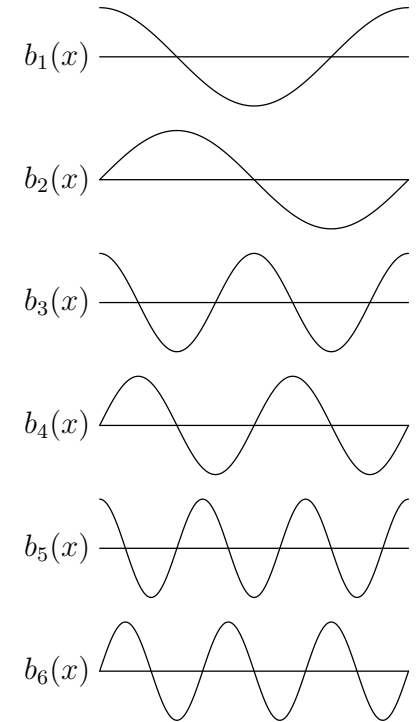
# Orthogonal Functions

- For functions we can use any ortho-normal set of functions as a basis
- The most familiar are the Fourier functions  $\sin(n\theta)$  and  $\cos(n\theta)$
- Any function in  $C(0,2\pi)$  can be represented by a point  $\mathbf{f} = \begin{pmatrix} \langle f(x), b_0(x) \rangle \\ \langle f(x), b_1(x) \rangle \\ \vdots \end{pmatrix}$
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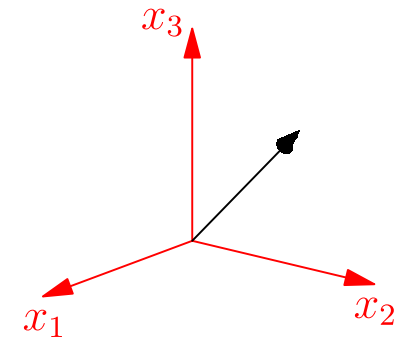
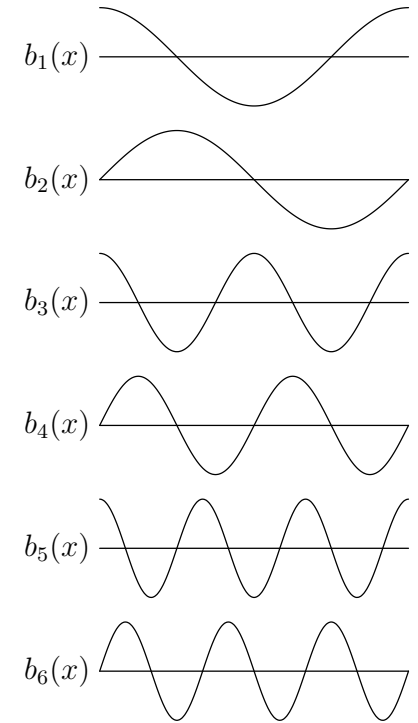
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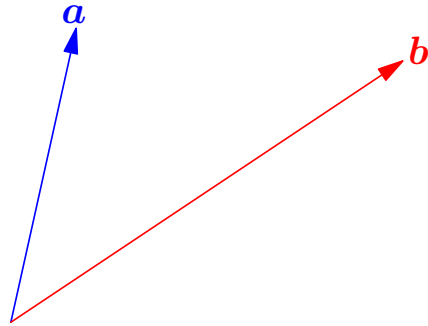
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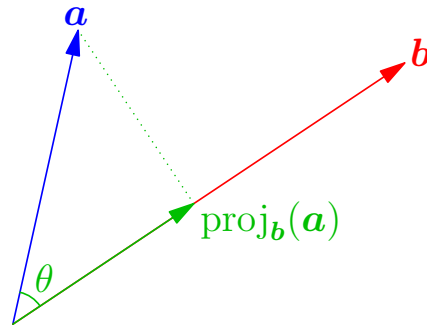
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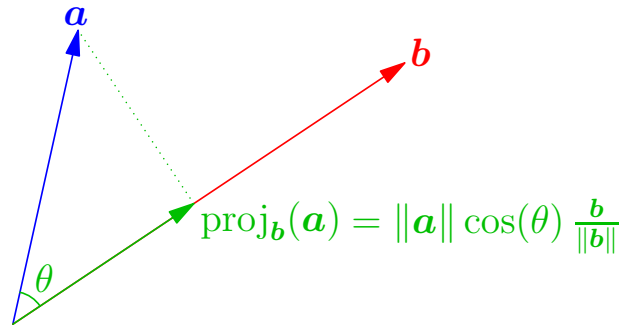
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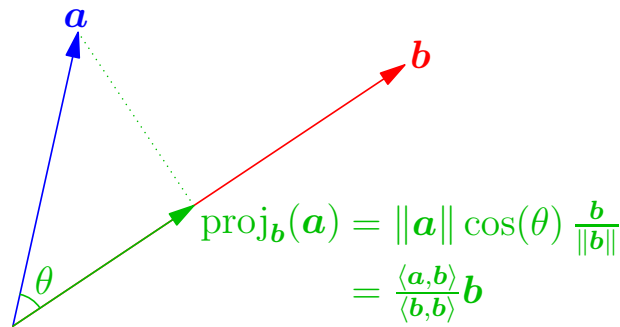
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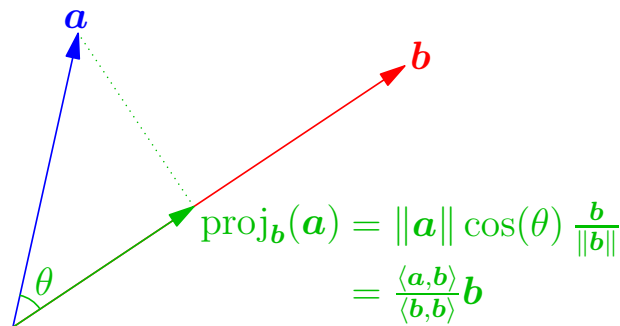
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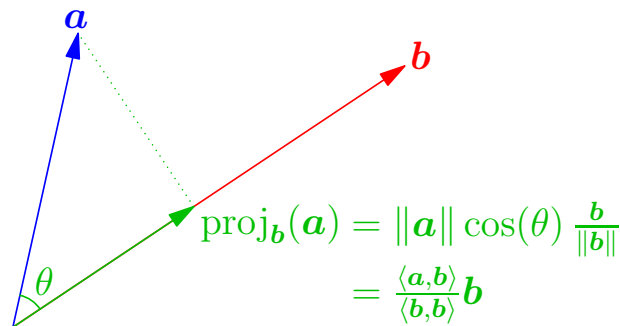


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$$\text{proj}_b(\text{proj}_b(a)) = \text{proj}_b\left(\frac{\langle a, b \rangle}{\langle b, b \rangle} b\right)$$

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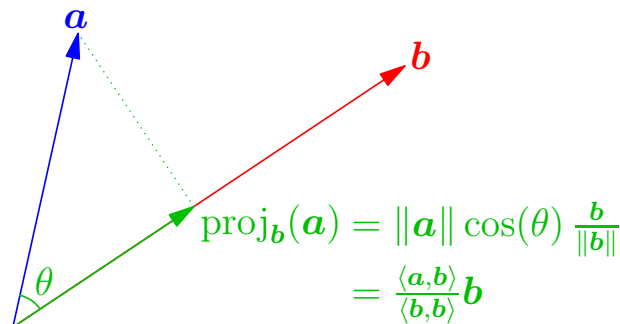


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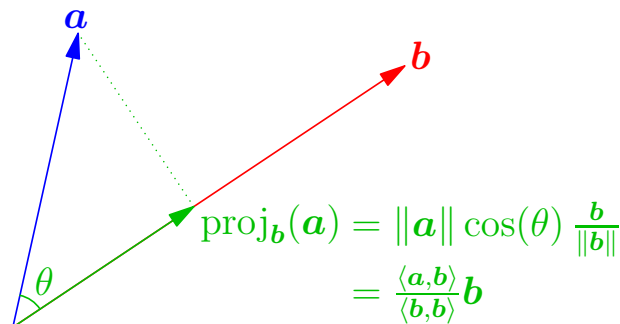


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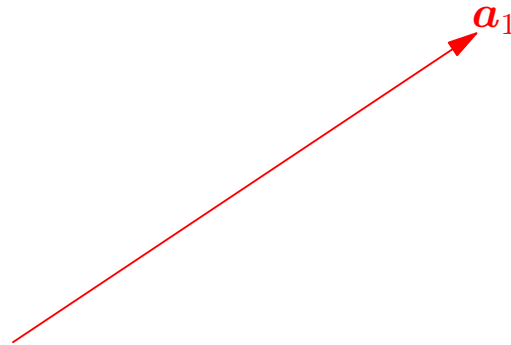
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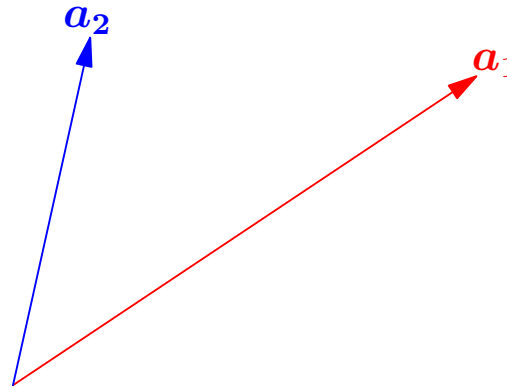
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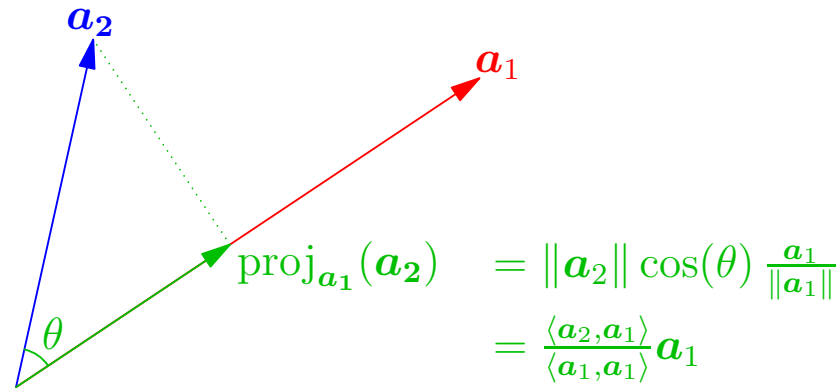
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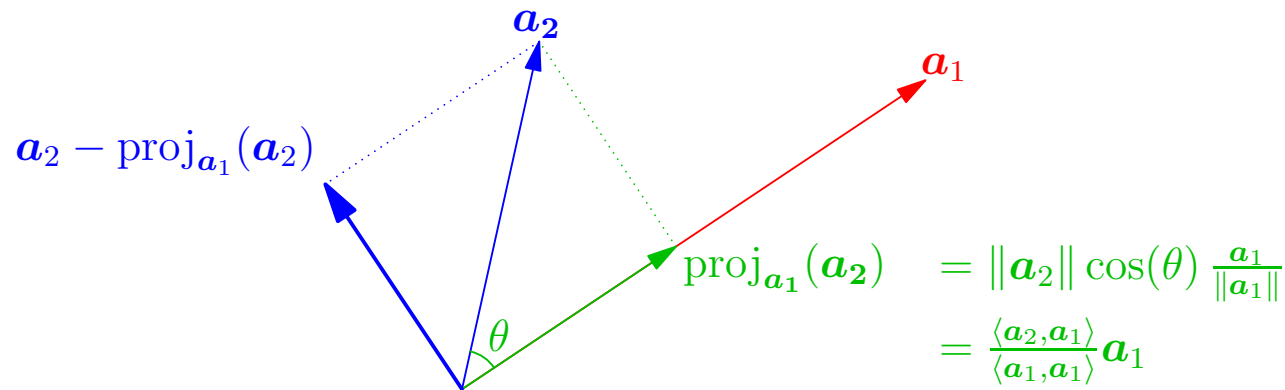
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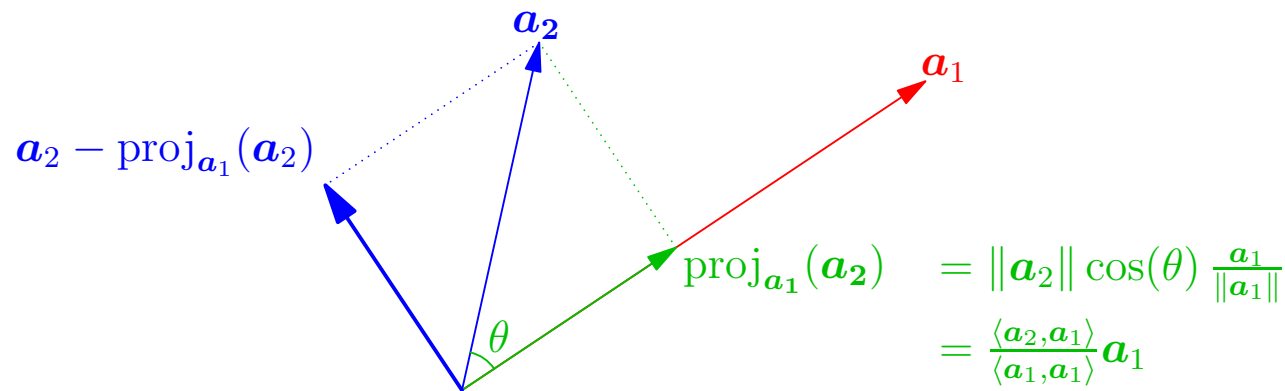
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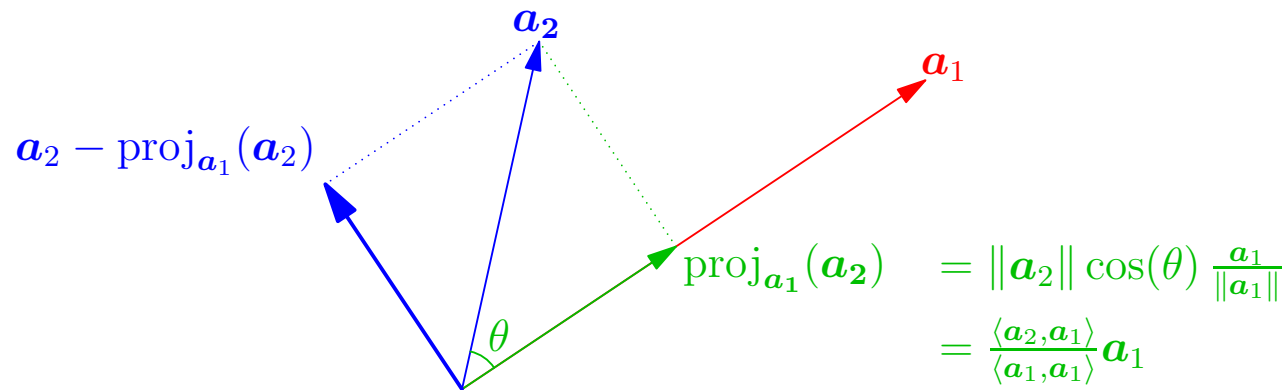
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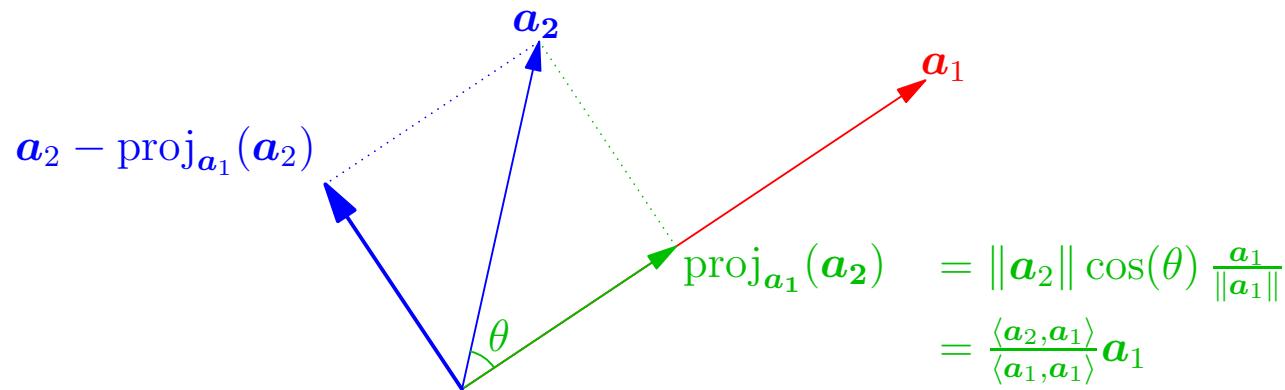
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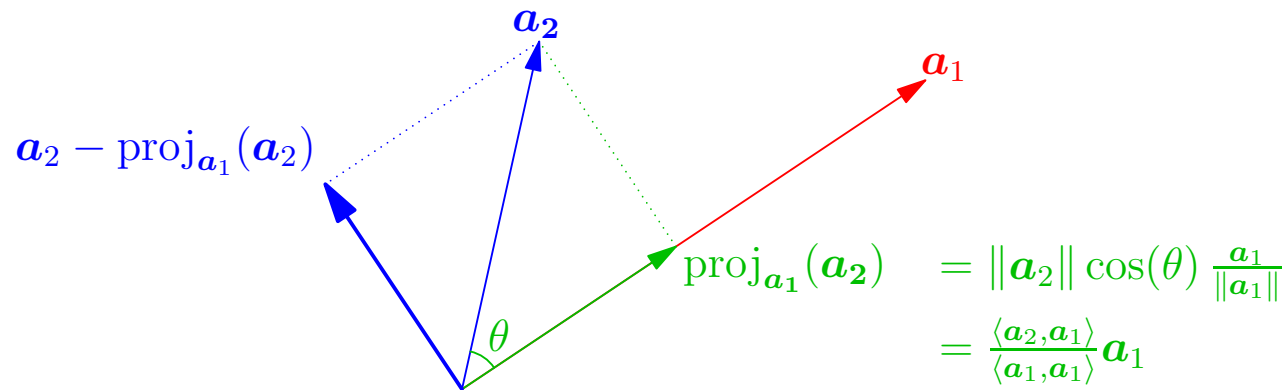
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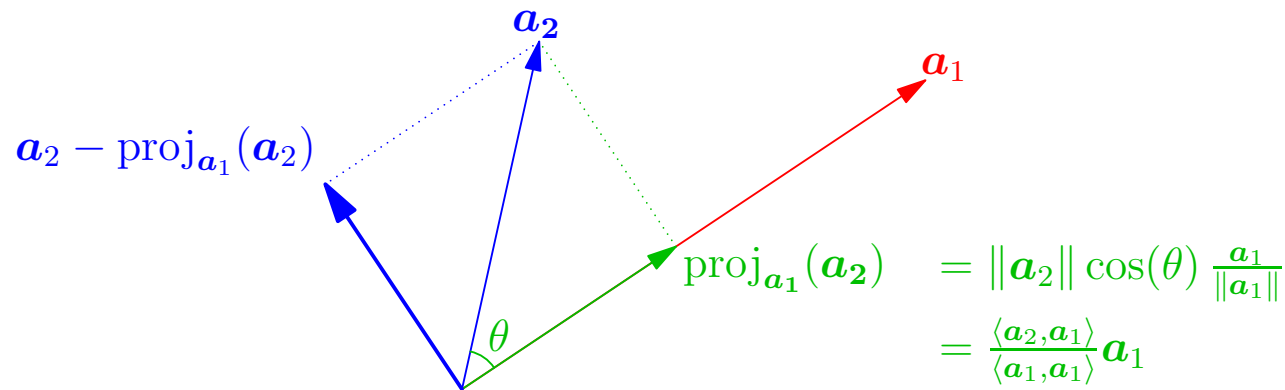
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- We have gone to these lengths as we want to show that many properties of vectors are shared by other objects (matrices, functions, etc.)
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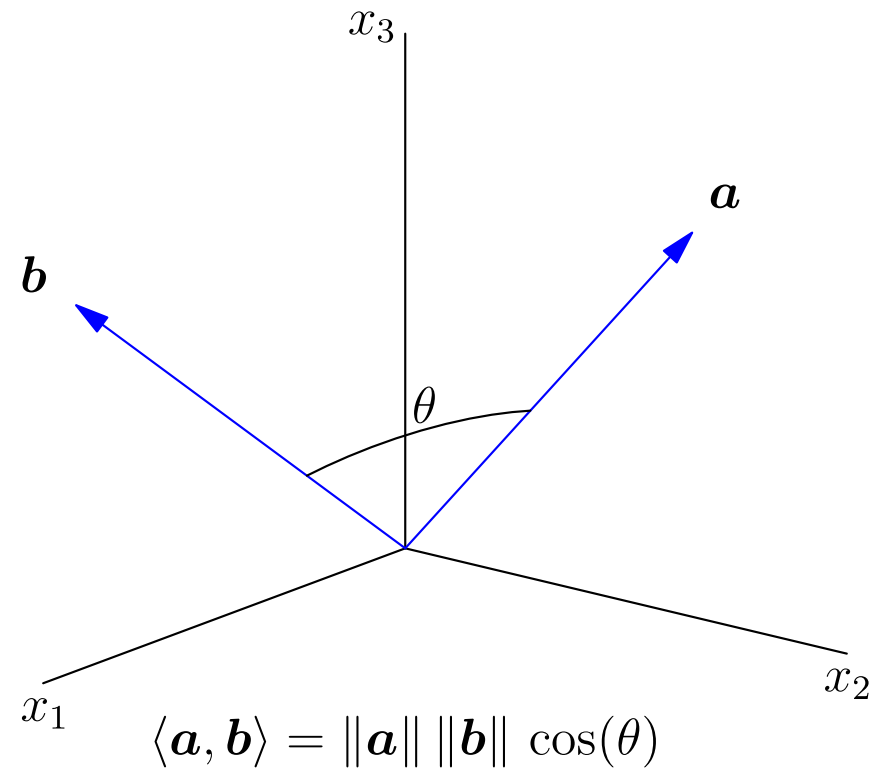
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# Outline

1. Inner Products
2. **Operators**



# Operators

- In machine learning we are interested in transforming our input vectors into some output predictions
- To accomplish this we will apply some mapping or operators on the vector  $\mathcal{T} : \mathcal{V} \rightarrow \mathcal{V}'$
- This says that  $\mathcal{T}$  maps some object  $x \in \mathcal{V}$  to an object  $y = \mathcal{T}[x]$  in a new vector space  $\mathcal{V}'$
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# Linear Operators

- Operators are in general very complicated, but a particular nice set of operators are linear operators
- $\mathcal{T}$  is a linear operator if
  1.  $\mathcal{T}[a\mathbf{x}] = a\mathcal{T}[\mathbf{x}]$
  2.  $\mathcal{T}[\mathbf{x} + \mathbf{y}] = \mathcal{T}[\mathbf{x}] + \mathcal{T}[\mathbf{y}]$
- For normal vectors ( $\mathbf{x} \in \mathbb{R}^n$ ) the most general linear operation is

$$\mathcal{T}[\mathbf{x}] = \mathbf{M}\mathbf{x}$$

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# Matrix multiplication

- For an  $\ell \times m$  matrix  $\mathbf{A}$  and an  $m \times n$  matrix  $\mathbf{B}$  we can compute the  $(\ell \times n)$  product,  $\mathbf{C} = \mathbf{AB}$ , such that

$$C_{ij} = \sum_{k=1}^m A_{ik} B_{kj} \quad \left( \begin{array}{|c|} \hline \text{---} \\ \hline \text{---} \\ \hline \text{---} \\ \hline \end{array} \right) \left( \begin{array}{|c|c|c|} \hline | & | & | \\ \hline | & | & | \\ \hline | & | & | \\ \hline \end{array} \right) = \left( \begin{array}{|c|c|c|} \hline \blacksquare & \blacksquare & \blacksquare \\ \hline \blacksquare & \blacksquare & \blacksquare \\ \hline \blacksquare & \blacksquare & \blacksquare \\ \hline \end{array} \right)$$

- Treating the vector  $\mathbf{x}$  as a  $n \times 1$  matrix then

$$\mathbf{y} = \mathbf{Ax} \quad \Rightarrow \quad y_i = \sum_j M_{ij} x_j \quad \left( \begin{array}{|c|} \hline \text{---} \\ \hline \text{---} \\ \hline \text{---} \\ \hline \end{array} \right) \left( \begin{array}{|c|} \hline | \\ \hline | \\ \hline | \\ \hline \end{array} \right) = \left( \begin{array}{|c|} \hline \blacksquare \\ \hline \blacksquare \\ \hline \blacksquare \\ \hline \end{array} \right)$$

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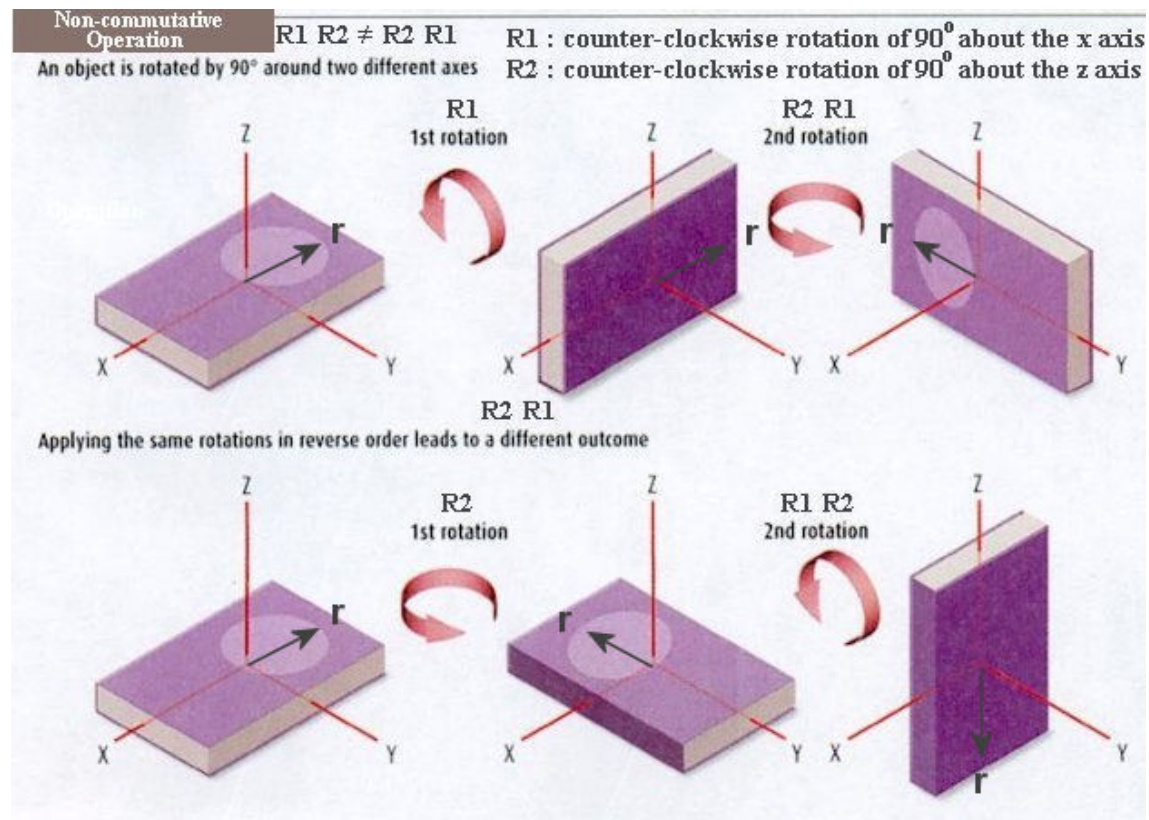
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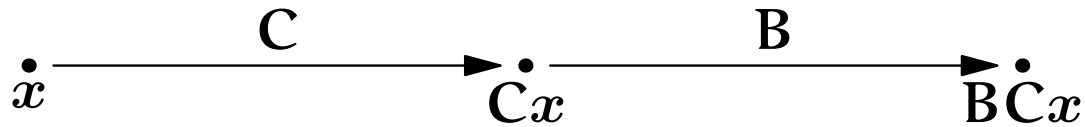
# Associativity of Mappings

$\dot{x}$

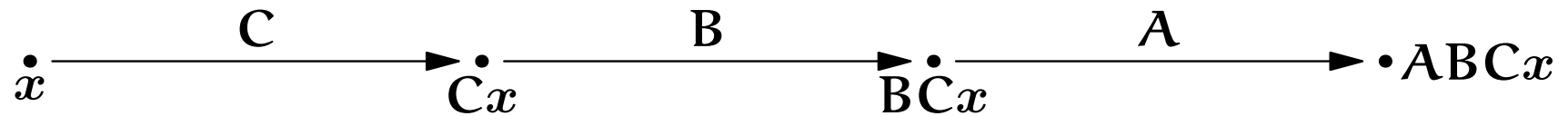
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$$\dot{x} \xrightarrow{C} \dot{C}x$$

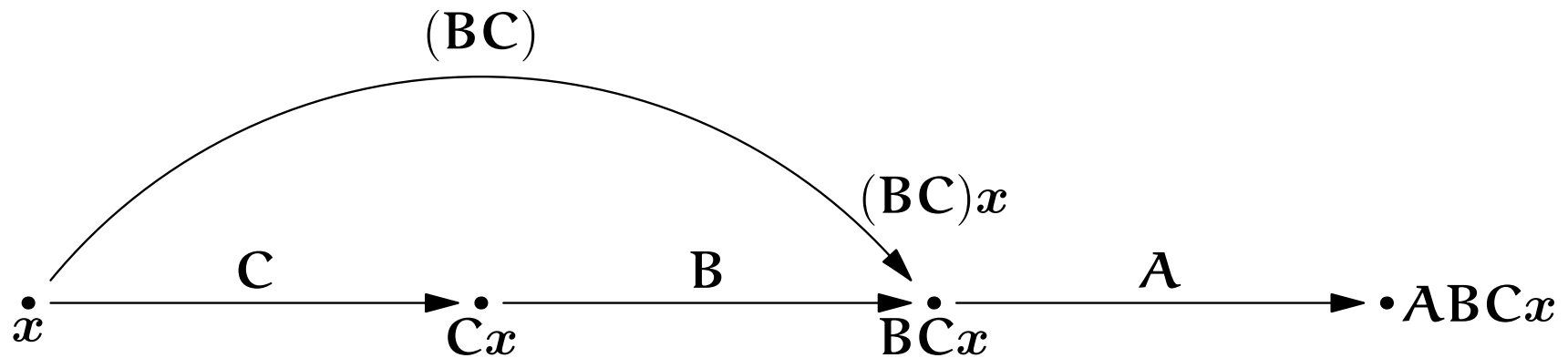
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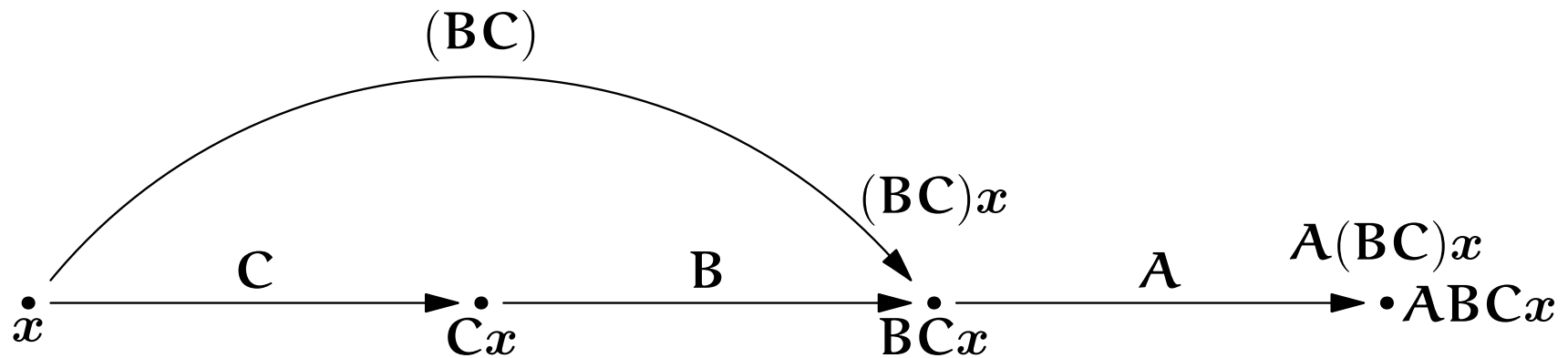
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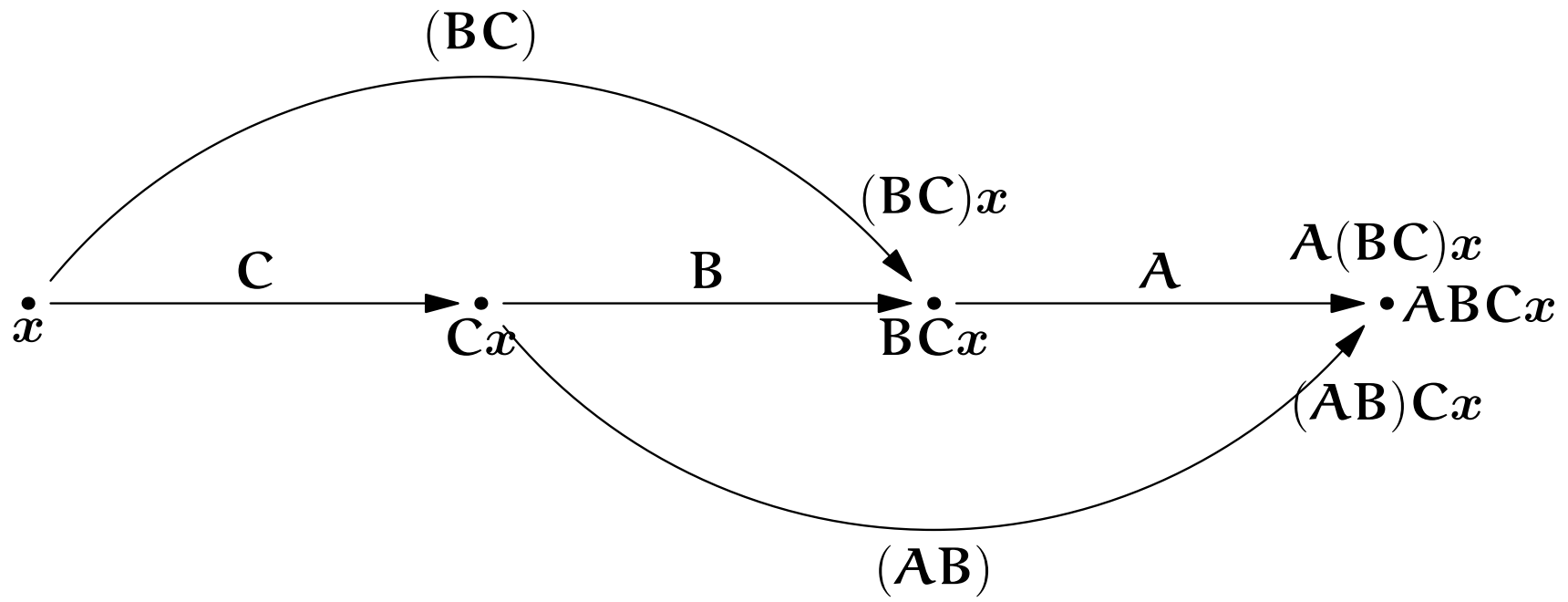
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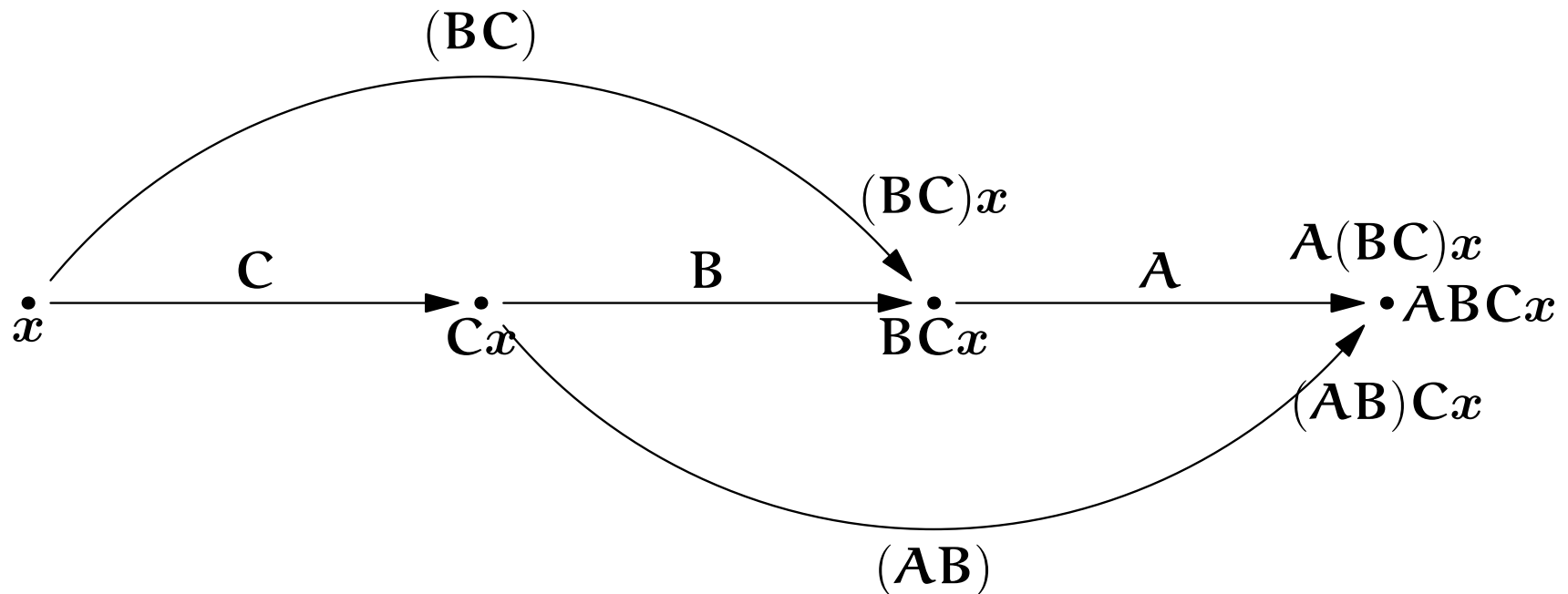
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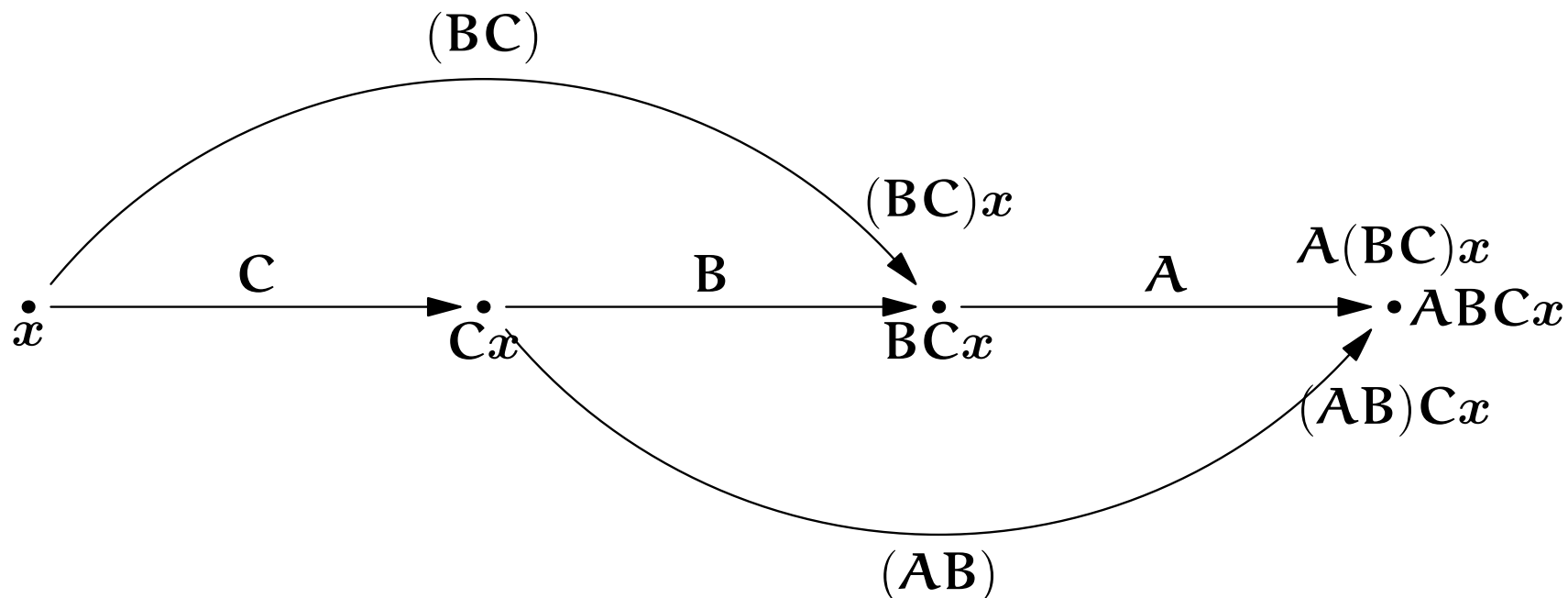


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# Kernels in Machine Learning

- Kernels are used heavily in machine learning
- In kernel methods such as SVM, SVR, Kernel-PCA
- They are also used in Gaussian Processes
- In all these cases we consider symmetric, positive semi-definite kernels
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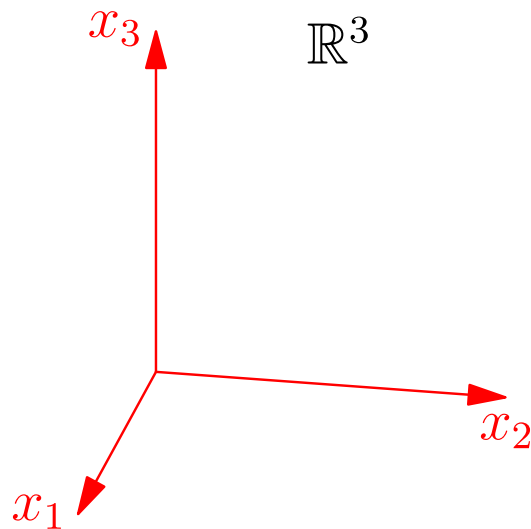
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- E.g.  $\mathbb{R}^3 \rightarrow \mathbb{R}^2$

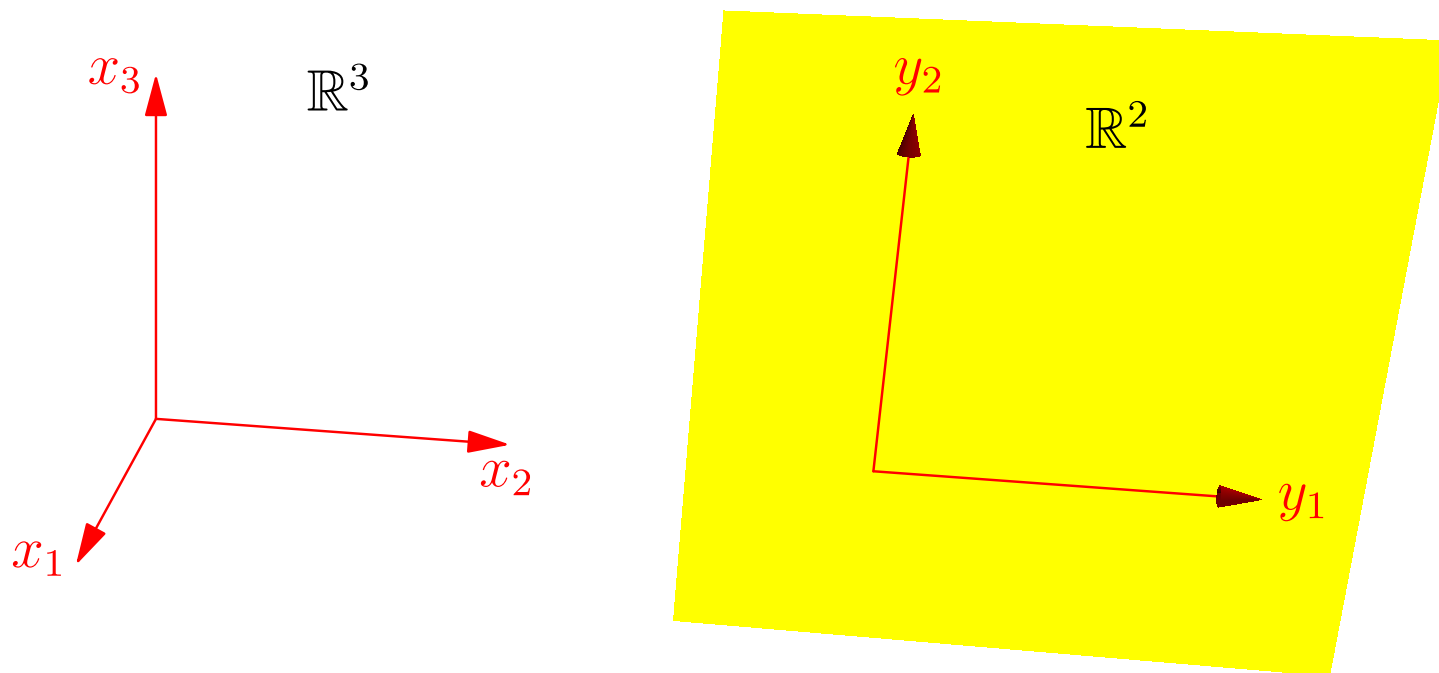
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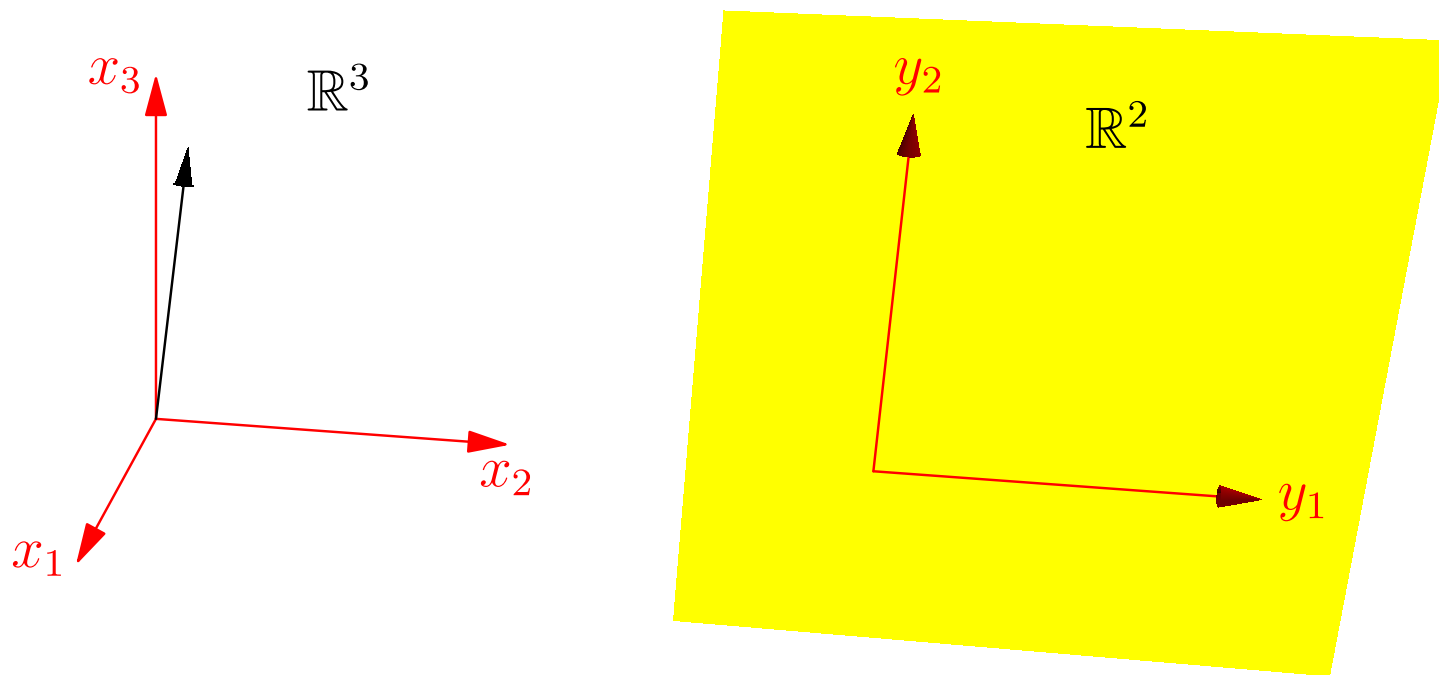
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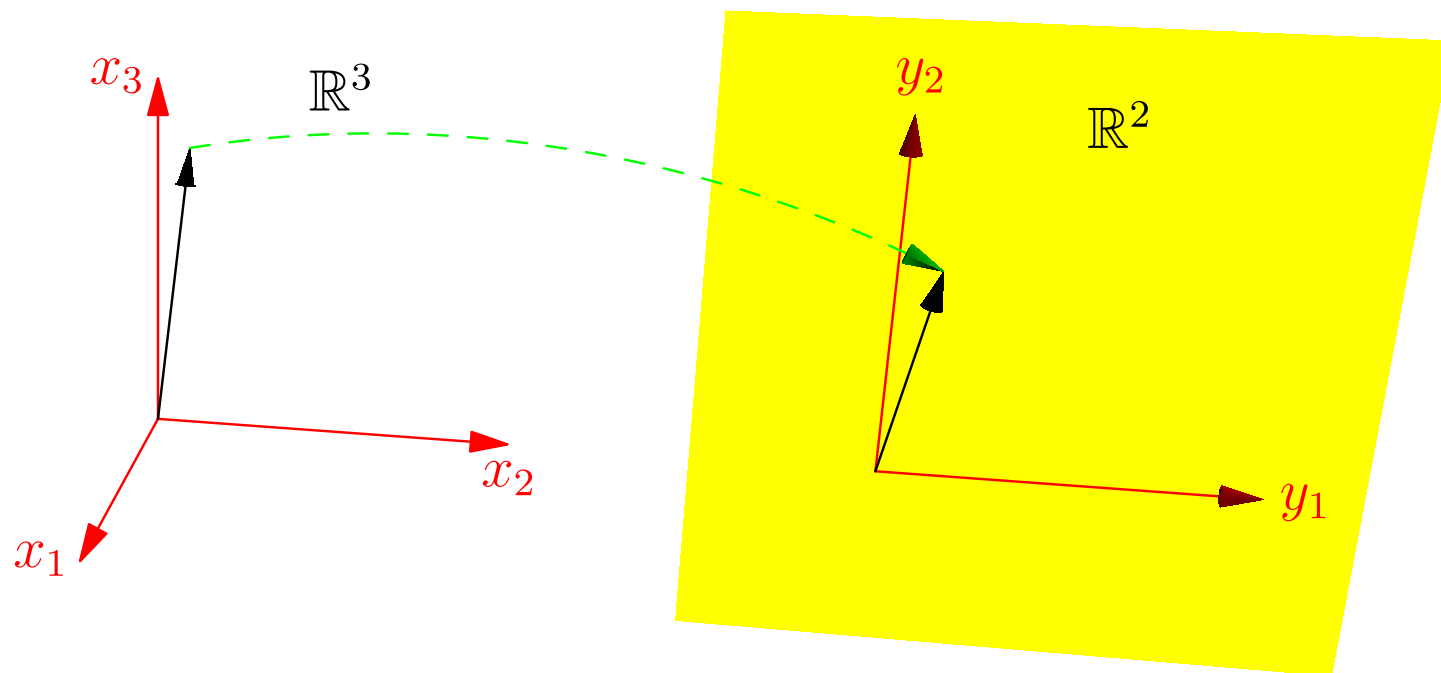
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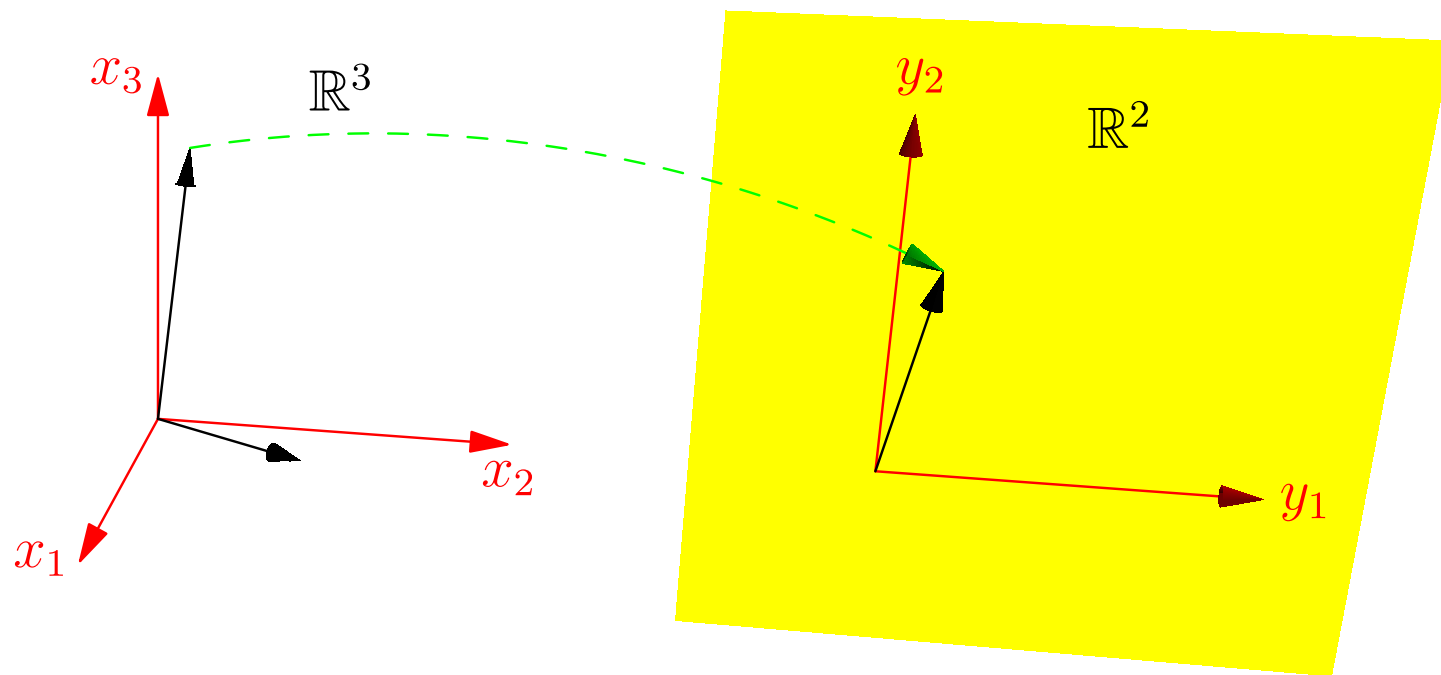
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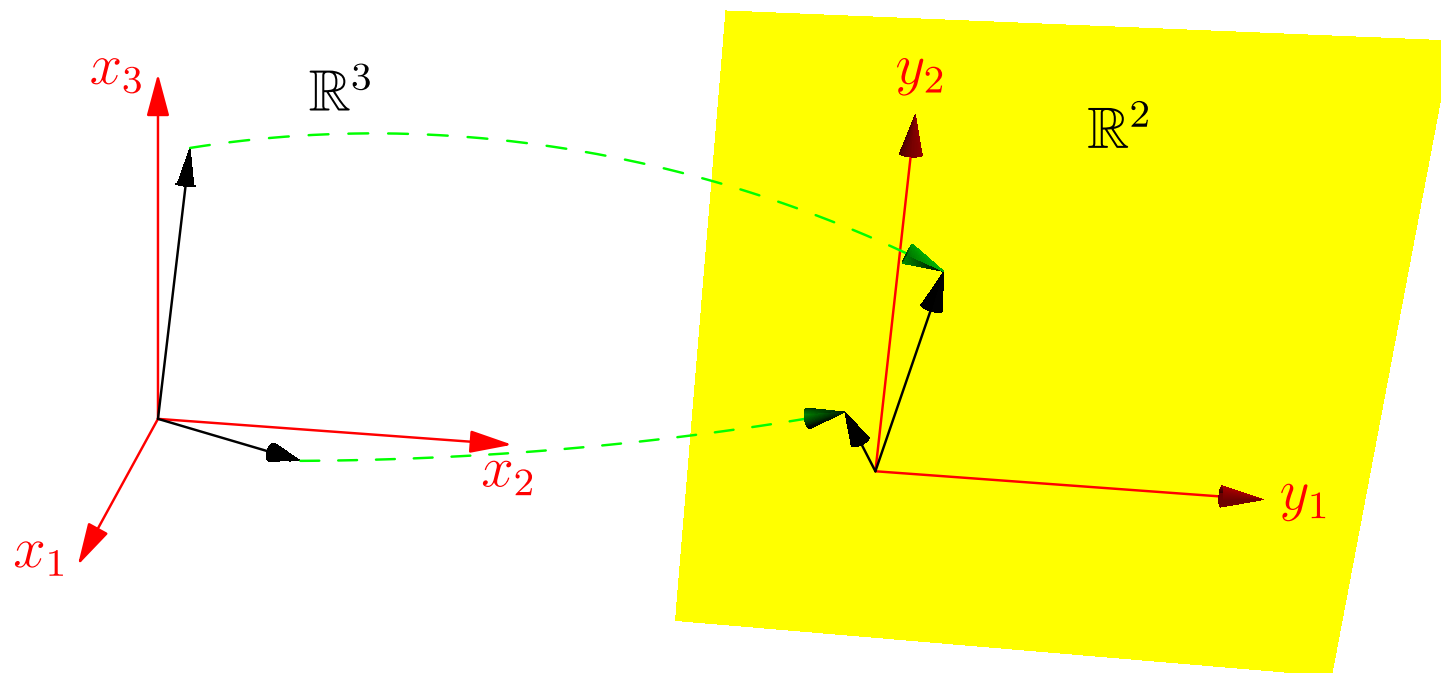
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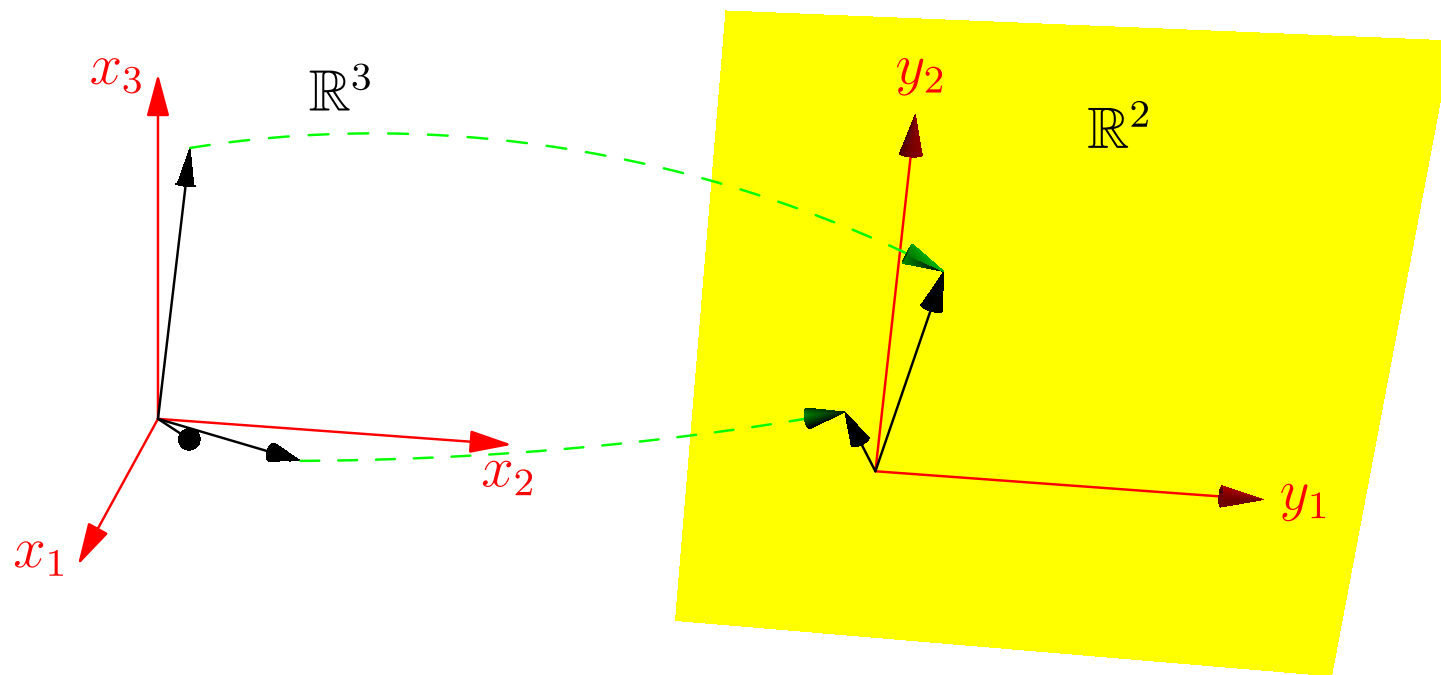
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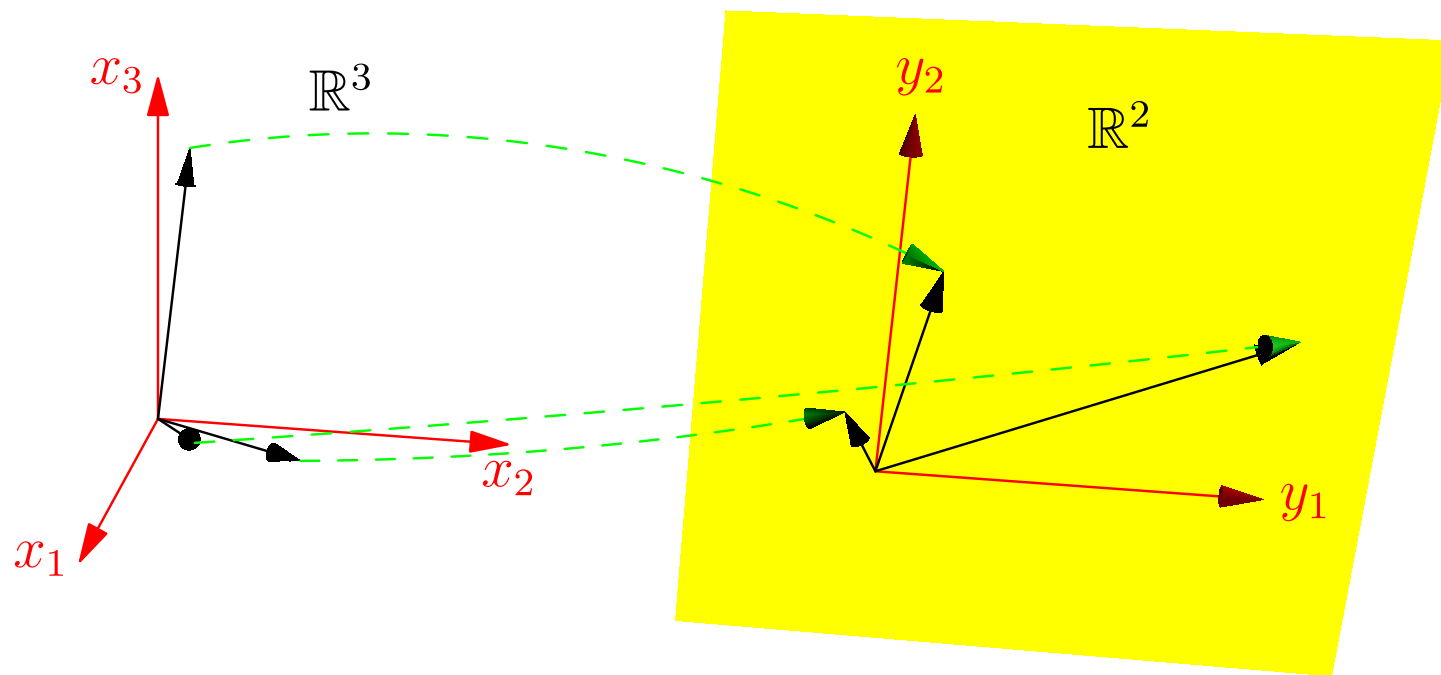
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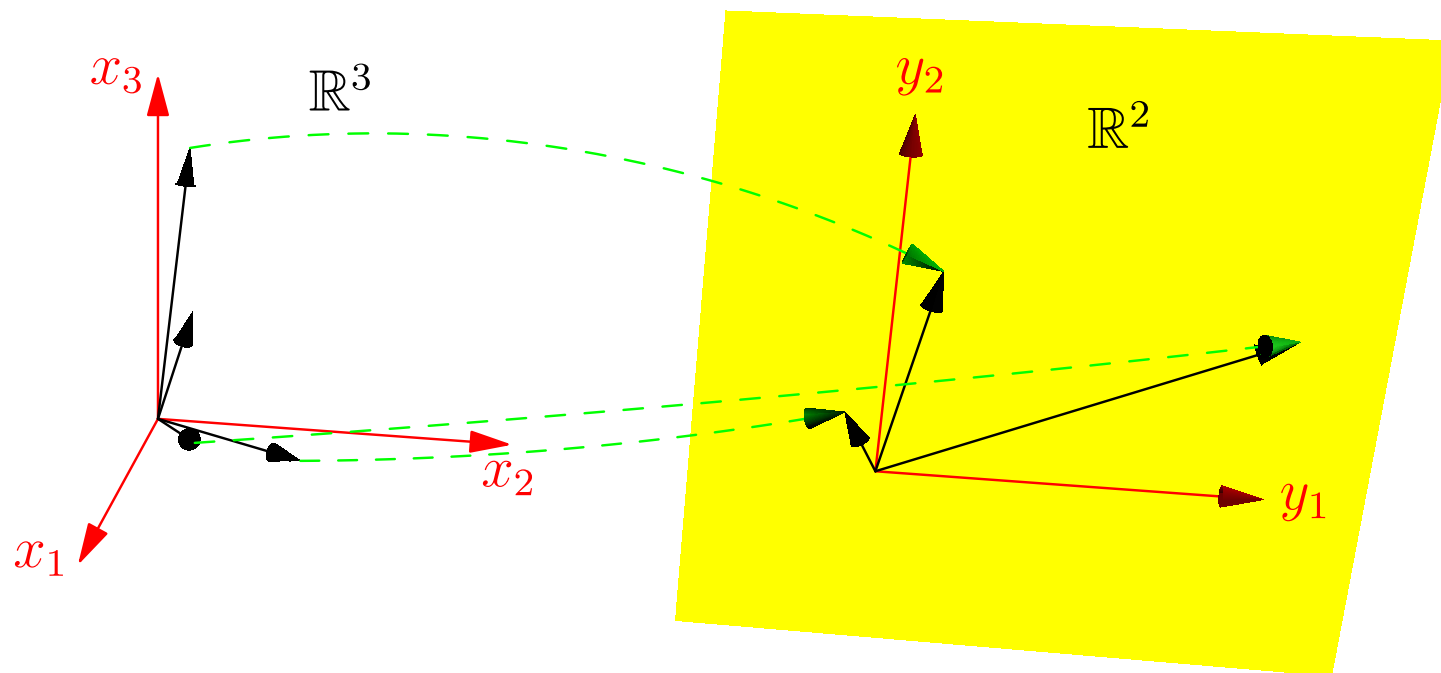
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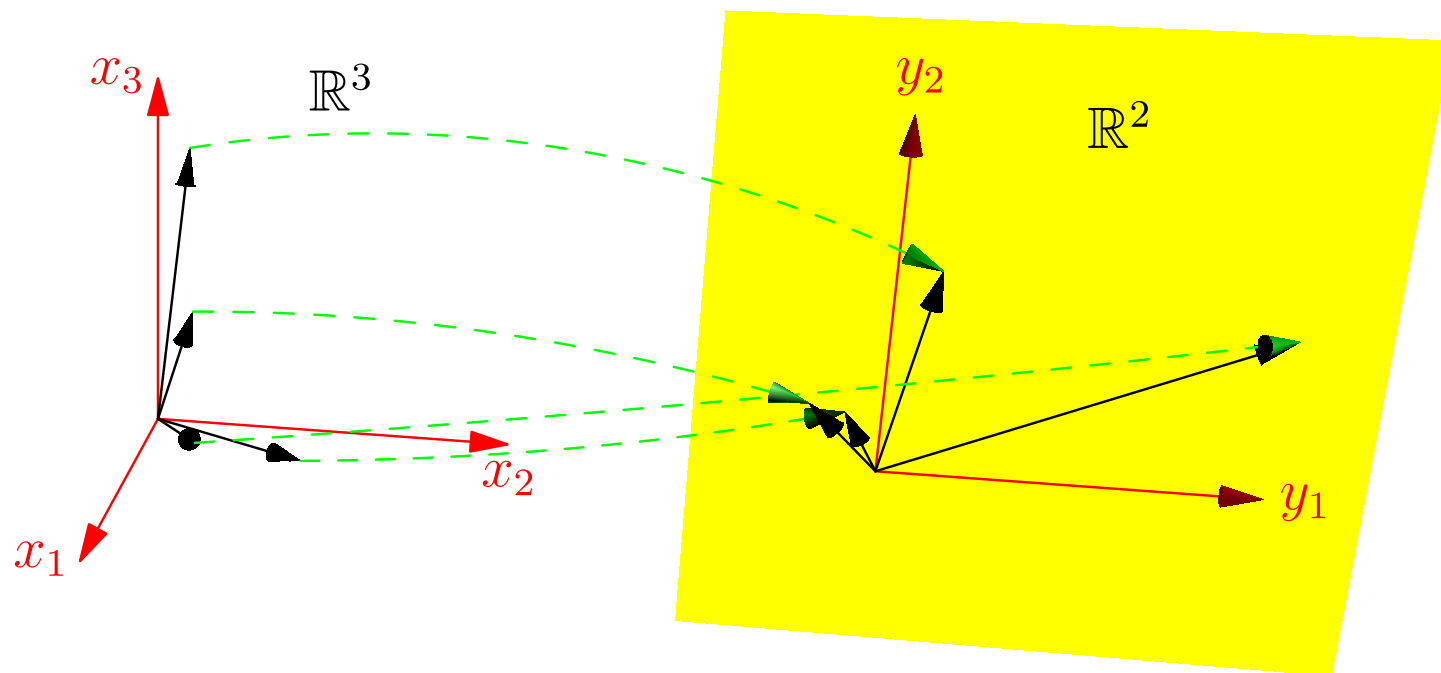
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